

The Relativistic Particle:

$$S = -m \int_{s_i}^{s_f} ds = -m \int_{t_0}^{t_f} dt \sqrt{-\frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \eta_{\mu\nu}}$$

i.e.

$$\gamma: \tau \mapsto X^\mu(\tau) \in \mathbb{M}^{2, d-1}$$

(Nambu-Goto action)

$$S \equiv \int d\tau L(\tau)$$

• If we identify τ w/ the

proper time:

$$-\frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \eta_{\mu\nu} = 1 - \vec{v}^2$$

$$S = -m \int d\tau \sqrt{1 - \vec{v}^2}$$

Recall that $X^\mu = (t, \vec{x})$

$$\frac{dt}{d\tau} = \gamma = (1 - \vec{v}^2)^{-1/2}$$

$$\text{and so } \frac{dX^\mu}{d\tau} = (\gamma, \gamma \vec{v}) = \gamma(1, \vec{v})$$

$$\therefore \left(\frac{dX^\mu}{d\tau}\right) \frac{dX^\nu}{d\tau} \eta_{\mu\nu} = -\gamma^2 + \vec{v}^2 \gamma^2 = -1$$

* Recall that the proper time is

$$-d\tau^2 = ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

Often, texts will use $\int ds$ when really

this is $\int d\tau$.

$$\Leftrightarrow d\tau/dt = \sqrt{-\eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu}$$

Since $X^\mu = (t, \vec{x})$

$$d\tau/dt = \sqrt{1 - \vec{v}^2}$$

$$\Leftrightarrow \boxed{\frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \vec{v}^2}} = \gamma}$$

i.e.

local

* Diffeomorphisms: let $\tau \rightarrow \tilde{\tau}(\tau)$

then $\frac{dx^\mu}{d\tilde{\tau}} = \frac{dx^\mu}{d\tau} \frac{d\tau}{d\tilde{\tau}}$

$$d\tilde{\tau} = \frac{\partial \tilde{\tau}}{\partial \tau} d\tau$$

$$\begin{aligned} \tilde{S} &= -m \int d\tilde{\tau} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tilde{\tau}} \frac{dx^\nu}{d\tilde{\tau}}} \\ &= -m \int \left(\frac{\partial \tilde{\tau}}{\partial \tau} d\tau \right) \sqrt{-\eta_{\mu\nu} \left(\frac{dx^\mu}{d\tau} \frac{\partial \tau}{\partial \tilde{\tau}} \right) \left(\frac{dx^\nu}{d\tau} \frac{\partial \tau}{\partial \tilde{\tau}} \right)} \\ &= -m \int d\tau \left(\frac{d\tilde{\tau}}{d\tau} \right) \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} \\ &= S \quad \checkmark \end{aligned}$$

* From this we are free to parametrize τ as we wish & we often choose affine coords.

Second order formalism: Introduce e or the Polyakov-action for massive ^{rel.} point particles in ND $e^2 \equiv g_{\tau\tau}$

$$S_p \equiv \frac{1}{2} \int d\tau e(\tau) \left(\frac{1}{e^2(\tau)} \left(\frac{dx^\mu}{d\tau} \right)^2 - m^2 \right)$$

• $\dot{x}^2 \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$

* Proof that $S_p = S_{NG}$:

$$\frac{\delta S_p}{\delta e} = 0 \Leftrightarrow -\frac{1}{e^2} (\dot{x})^2 - m^2 = 0$$

$$e^2 = -\frac{\dot{x}^2}{m^2}$$

$$\text{and } \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu(\tau)} = 0 \Rightarrow g_{\mu\nu} \dot{x}^\nu = \frac{d}{d\tau} \left(\sum_{\alpha} \delta(\tau - \tau') \right) \frac{dx^\mu}{d\tau} \\ - \int d\tau' 2 \delta(\tau - \tau') \frac{d}{d\tau} (e^{-1}(\tau) \dot{x}^\mu) = 0 \\ \boxed{\frac{d}{d\tau} (e^{-1} \dot{x}^\mu) = 0}$$

Since E.O.M of e are algebraic it has no dynamics and is purely aux.

The Canonical Momenta: $P^\mu := \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

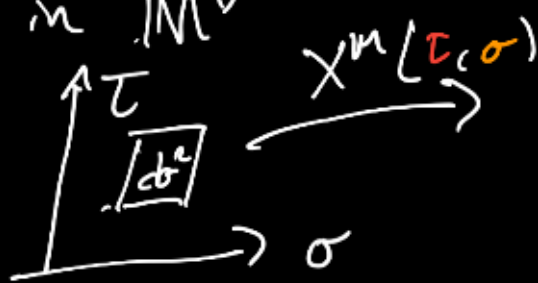
Here $P^\mu = \dot{x}^\mu(\tau)$

For $e^2 = -\dot{x}^2/m^2$ $|e| = 1$ we simply have the mass-shell condition

$$\boxed{P_\mu P^\mu + m^2 = 0}$$

The Bosonic String:

- We can, topologically, only have open or closed strings. These will be embedded in M^D



Σ : World sheet

Which is parametrized by $\tau \in \mathbb{R}, 0 \leq \sigma \leq l$

$$\Sigma: (\tau, \sigma) \mapsto X^\mu(\tau, \sigma) \in \mathbb{M}^D$$

which we define to be the string map.

$$\Sigma \hookrightarrow \mathbb{M}^D$$

We use this to find the action of the string. Above, we calculated the minima of the length of γ defined by a particle. Here we minimize the Area of the string.

Let $\sigma^a \equiv (\tau, \sigma)$ be worldsheet coords
 $\sigma^0 = \tau, \quad \sigma^1 = \sigma$

then we know

$$X: \Sigma \rightarrow \mathbb{M}^D$$

$$(\tilde{X})^*: T^*\mathbb{M}^D \rightarrow T^*\Sigma$$

$$\begin{aligned} (\tilde{X})^* g(X, Y) &= g(\tilde{X}_* X, \tilde{X}_* Y) \\ &= g(\partial_\alpha \tilde{X}^\mu \partial_\mu, \partial_\beta \tilde{X}^\nu \partial_\nu) \\ \Leftrightarrow (\tilde{X}^* g)_{\alpha\beta} &= \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \end{aligned}$$

$\therefore$ The metric on Σ is given by

$$G_{\alpha\beta}: T\Sigma \times T\Sigma \rightarrow \mathbb{R}$$

$$G_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X^\nu$$

$$S_{NG} = -T \int dA = -T \int \sqrt{\det h_{\alpha\beta}} d^2\sigma$$

$$G_{00} = \dot{X}^2$$

$$G_{11} = \frac{dX^\mu}{d\sigma} \frac{dX^\nu}{d\sigma} \equiv \dot{X}'^2$$

$$G_{01} = \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\sigma} \eta_{\mu\nu}$$

$$[G_{\alpha\beta}] = \begin{pmatrix} \dot{x}^2 & \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\tau} \eta_{\mu\nu} \\ \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\sigma} \eta_{\mu\nu} & x'^2 \end{pmatrix}$$

$$\det(\) = \dot{x}^2 x'^2 - (\dot{x} \cdot x')^2$$

$$S_{\text{NG}} = -\frac{T}{2} \int d\tau d\sigma \sqrt{(\dot{x} \cdot x')^2 - (\dot{x}^2) (x')^2}$$

* We introduce the auxiliary field, worldsheet metric $h_{\alpha\beta}(\tau, \sigma)$ which is different from the pull back $G_{\alpha\beta}$

Polyakov Action:

$$\begin{aligned} S_P &\equiv -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta}(\sigma) \partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu} \\ &= -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} G_{\alpha\beta} \end{aligned}$$

* For a spacetime M^D w/ metric $g_{\mu\nu}$ we simply replace $\eta_{\mu\nu}$ w/ $g_{\mu\nu}(x)$. This leads to a non-linear sigma-model.

$$\rightarrow g_{\mu\nu}(x) = \eta_{\mu\nu}$$

Can be interpreted as a lowest order

$$\text{expansion } g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + \mathcal{O}(h^2)$$

$$\delta S_P = -\frac{T}{2} \int d^2\sigma \delta(\sqrt{-h} h^{\alpha\beta}) \partial_\alpha x^\mu \partial_\beta x^\nu g_{\mu\nu}$$

$$= -\frac{T}{2} \int d^2\sigma \left[\left(-\frac{1}{2} \sqrt{-h} h^{\alpha\beta} \delta h^{\alpha\beta} \right) h^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x^\nu g_{\mu\nu} \right]$$

$$= -\frac{1}{2} \int d^2\sigma (\delta h^{\alpha\beta}) \sqrt{-h} \left[\frac{1}{2} h_{\alpha\beta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \right]$$

The energy-momentum tensor:

$$T_{\alpha\beta} := \left(-\frac{2}{\sqrt{-h}} \right) \frac{\delta S}{\delta h^{\alpha\beta}}$$

$$= -\frac{1}{2} \int d^2\sigma \sqrt{-h} (\delta h^{\alpha\beta}) \left(\frac{1}{2} h_{\alpha\beta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} + \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \right)$$

$$T_{\alpha\beta} = -\frac{1}{2} h_{\alpha\beta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} + \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}$$

Note to self: In $\delta \sqrt{-h}$ recall

that the $-$ must be reabsorbed in order to have $(h)/\sqrt{-h} = \sqrt{-h}$! i.e.

$$\begin{aligned} \delta \sqrt{-h} &= \left(-\frac{1}{2} \right) \frac{1}{\sqrt{-h}} \delta h = \left(\frac{1}{2} \frac{1}{\sqrt{-h}} \right) (-h) (\delta h^{\alpha\beta}) \\ &= -\frac{1}{2} \sqrt{-h} h_{\alpha\beta} \delta h^{\alpha\beta} \end{aligned}$$

*E.O.M for $h^{\alpha\beta}$, i.e. (second order formalism):

$$T_{\alpha\beta} = 0 \Leftrightarrow \frac{1}{2} h_{\alpha\beta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} = \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}$$

• On-shell, the pull-back metric $\rightarrow$ and the induced metric $h_{\alpha\beta}$ are proportional.

$$G_{\alpha\beta} = h_{\alpha\beta} \left(\frac{1}{2} h^{\alpha\beta} G_{\alpha\beta} \right)$$

*E.O.M for X^μ :

$$\delta S = -\frac{1}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha \delta X^\mu \partial_\beta X^\mu$$

$$= -\frac{1}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha \delta X^\mu \partial_\beta X^\mu$$

$$\Rightarrow \Delta x^\mu = \frac{1}{\sqrt{-h}} \partial_\alpha (\sqrt{-h} h^{\alpha\beta} \partial_\beta x^\mu) = 0$$

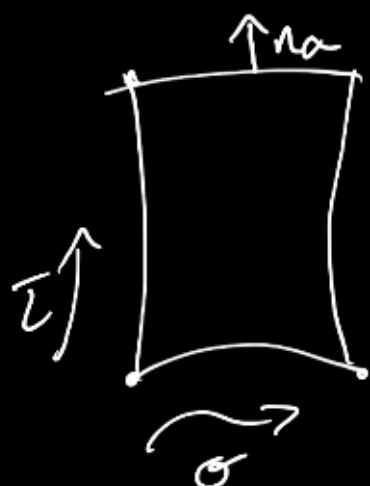
which has B.C.'s:

(i) Closed String:

$$x^\mu(\tau, \sigma + \ell) = x^\mu(\tau, \sigma)$$

(ii) Open string:

$$\eta^{\alpha\beta} \partial_\alpha x^\mu \delta x_\mu |_{\sigma=0,\ell} = 0$$



$T_{\alpha\beta}$ is covariantly constant:

$$\nabla_\alpha T^{\alpha\beta} = 0$$

* The Polyakov metric is dot, like NG action, the Area of the worldsheet.

$$\text{Area: } \int d^2\sigma \sqrt{-h}$$

Let A be a symm. $\in M_{2 \times 2}(\mathbb{R})$. Then

$$(\text{tr}(A))^2 \geq 4 \det(A)$$

$$\text{Proof: } \text{Tr}(A) = \sum_{i=1}^n \lambda_i, \det(A) = \prod_{i=1}^n \lambda_i$$

then AM-GM inequality states

$$\frac{x_1 + \dots + x_n}{n} \geq (x_1 \dots x_n)^{1/n}$$

$$\Rightarrow A \in M_{2 \times 2}(\mathbb{R}) \Leftrightarrow n = 2$$

$$\left(\frac{\lambda_1 + \lambda_2}{2} \right) \geq (\lambda_1 \lambda_2)^{1/2}, \text{ or}$$

$$(\lambda_1 + \lambda_2)^2 \geq 4 \lambda_1 \lambda_2$$

$$\text{Tr}(A)^2 \geq 4 \det(A)$$

Equality holds for $A \propto \mathbb{I}$.

* For $A^\alpha_\beta = h^\alpha_\gamma G_{\gamma\beta} \Leftrightarrow S_p \geq S_{NG}$.

Equality holds when $G_{\gamma\beta}$ is conformally eq.

to $h_{\alpha\beta}$. $\Leftrightarrow$ On-shell $T_{\alpha\beta} = 0$ condition implies conformal equivalence of the pull back and $h_{\alpha\beta}$.

Q: Are there other terms which we could add to the string action?

(i) Cosmological constant term can be added

$$S_1 = \Lambda_2 \int d^2\sigma \sqrt{-h}$$

• We must have Λ_2 to have Weyl invariance.

(ii) 2D Einstein-Hilbert term

$$S_2 = \frac{\Lambda_2}{4\pi} \int d^2\sigma \sqrt{-h} R^{(2)}$$

This has no dynamics as 2D gravity is trivial

$$\chi(\Sigma) = \frac{1}{4\pi} \int d^2\sigma \sqrt{-h} R^{(2)} \quad (\text{Euler characteristic})$$

(For closed string)

$$R^{(2)} \sqrt{-h} \sim \partial_\alpha (\quad)$$

The Polyakov action for propagation of a string

in $1d$ describes d two dimensional scalar fields $x^m(\sigma)$ coupled to the dynamical WS metric $h_{\alpha\beta}(\sigma)$

→ Studying Bosonic ST is equivalent to studying 2-d gravity coupled to scalars

Polyakov symmetries:

We must distinguish between Spacetime on M^D and the WS symm. acting on Σ .

i) Spacetime symmetries:

We have Poincaré invariance

$$x^m(\sigma) \mapsto \Lambda^m_n (x^n - a^n), \quad \Lambda \in SO(1, d-1)$$

Recall: $(\eta_{\mu\nu})^m_n := \epsilon^m_\rho \eta_{\mu\rho} - \delta^m_\mu \eta_{\rho\nu}$

$$\Leftrightarrow \delta x^m = \Lambda^m_n x^n + a^m$$

$$\delta h_{\alpha\beta} = 0$$

$$\text{Diff}(\Sigma_g) \equiv \{f: \Sigma_g \rightarrow \Sigma_g\}$$
$$f^* x^m$$

$$f^* h_{\alpha\beta} = \frac{\partial x^\mu}{\partial \xi^\alpha} \frac{\partial x^\nu}{\partial \xi^\beta} h_{\mu\nu}$$

(local) WS symmetries:

→ Reparametrization invariance

$$\sigma^\alpha \mapsto \sigma'^\alpha := \sigma^\alpha - \xi^\alpha(\sigma)$$

i) Scalar: $x^m(\sigma) \mapsto x'^m(\sigma') = x^m(\sigma(\sigma'))$

$$x'^m(\sigma') = x^m(\sigma') + \xi^\alpha \partial_\alpha x^m|_\sigma$$

$$\Rightarrow \delta x^m = x'^m(\sigma) - x^m(\sigma) = \xi^\alpha \partial_\alpha x^m(\sigma)$$

(ii) Metric:

$$\delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha$$

(iii) $\sqrt{-h}$:

$$\delta \sqrt{-h} = \frac{1}{2} h^{\alpha\beta} (\xi^\gamma \partial_\gamma h_{\alpha\beta} + \partial_\alpha \xi^\gamma h_{\gamma\beta} + \partial_\beta \xi^\gamma h_{\alpha\gamma})$$

Note $\partial_\alpha \sqrt{-h} = -\frac{1}{2} \frac{1}{\sqrt{-h}} h^{\gamma\delta} \partial_\alpha h_{\gamma\delta}$

$$\Rightarrow \delta \sqrt{-h} = \partial_\alpha (\xi^\alpha \sqrt{-h})$$

* For $\Lambda^\mu_\nu = \delta^\mu_\nu + \lambda^\mu_\nu$

$$\Rightarrow \delta x^\mu = \lambda^\mu_\nu x^\nu$$

To see that λ is antisymmetric:

$$\eta_{\mu\nu} x^\mu x^\nu = \eta_{\mu\nu} x'^\mu x'^\nu$$

$$\Rightarrow \delta(\eta_{\mu\nu} x^\mu x^\nu) = 0$$

$$2 \eta_{\mu\nu} \lambda^\mu_\sigma x^\sigma x^\nu = 0$$

$$2 \underbrace{\lambda_{\nu\sigma}}_{\text{Antisym}} \underbrace{x^\sigma x^\nu}_{\text{Sym}} = 0$$

examples of Poincare invariance:

(i) Rotation about x^3 by an angle θ

$$x'^1 = \cos(\theta) x^1 + \sin(\theta) x^2$$

$$x'^2 = -\sin(\theta) x^1 + \cos(\theta) x^2$$

$$\theta \ll 1, \cos \rightarrow 1, \sin \rightarrow \theta$$

$$x'^1 = x^1 + \theta x^2$$

$$x'^2 = -\theta x^1 + x^2$$

$$\delta x^1 = \theta x^2, \quad \delta x^2 = -\theta x^1$$

but since $\delta x^m = a^m_n x^n$

$$a^1_2 = \theta, \quad a^2_1 = -\theta \Rightarrow a_{12} = \eta_{11} a^1_2 = \theta$$

(ii) Boost in the x^0 & x^1 direction by φ .

$$x'^0 = \cosh(\varphi) x^0 + \sinh(\varphi) x^1$$

$$x'^1 = \sinh(\varphi) x^0 + \cosh(\varphi) x^1$$

$$\cosh \rightarrow 1, \quad \sinh \rightarrow \varphi$$

$$\delta x^0 = \varphi x^1, \quad \delta x^1 = \varphi x^0$$

$$\Rightarrow \lambda^0_1 = \varphi, \quad \lambda^1_0 = \varphi$$

Invariance of S_p under Poincaré:

$$\begin{aligned} \delta S &= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha (\delta x^m) \partial_\beta x^n g_{mn} \\ &= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha (\lambda^m_\sigma x^\sigma + a^m) \partial_\beta x^n g_{mn} \\ &= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \lambda^m_\sigma \partial_\alpha x^\sigma \partial_\beta x^n g_{mn} \\ &= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \underbrace{\lambda_{\sigma\alpha}}_{\text{Anti}} \underbrace{\partial_\alpha x^\sigma \partial_\beta x^\nu}_{\text{Sym}} g_{\mu\nu} \end{aligned}$$

$\therefore \delta S = 0$ (Poincaré invariant)

* World Sheet (local) Symmetries:

$$\boxed{h_{\alpha\beta} \mapsto h'_{\alpha\beta} \equiv e^{2\phi(\sigma)} h_{\alpha\beta}(\sigma)}$$

$$\Rightarrow \delta h_{\alpha\beta} = 2\phi(\sigma) h_{\alpha\beta}$$

$$\delta x^m = 0 \quad \forall m.$$

(i) Note that $\sqrt{-h}$ is a tensor density

and transforms exactly as $h_{\alpha\beta}$

$$\sqrt{-h} = \sqrt{-\det(e^{2\phi} h_{\alpha\beta})} = \sqrt{-e^{4\phi} h}$$

$$= e^{2\phi} \sqrt{-h}$$

* The inverse of h transforms as

$$h^{\alpha\beta} \rightarrow e^{-2\phi} h^{\alpha\beta}$$

The pair $\sqrt{-h} h^{\alpha\beta}$ has conformal weight 0:

$$\sqrt{-h} h'^{\alpha\beta} = e^{2\phi} \sqrt{-h} e^{-2\phi} h^{\alpha\beta} = \sqrt{-h} h^{\alpha\beta} \checkmark$$

Claim: Conf. invariance implies the E.M tensor is traceless: $h^{\alpha\beta} T_{\alpha\beta} = 0$

Proof: $T_{\alpha\beta} = -\frac{2}{T} \frac{1}{\sqrt{-h}} \frac{\delta S}{\delta h^{\alpha\beta}}$, thus

$$\begin{aligned} \delta S &= \int \frac{\delta S}{\delta h^{\alpha\beta}} \delta h^{\alpha\beta} d^2\sigma \\ &= -\frac{T}{2} \int d^2\sigma \sqrt{-h} T_{\alpha\beta} \delta h^{\alpha\beta} \\ &= T \int d^2\sigma \sqrt{-h} T_{\alpha\beta}(\phi) h^{\alpha\beta} \stackrel{!}{=} 0 \end{aligned}$$

Since $\phi \neq 0$ and $\sqrt{-h} \geq 0$ we find

$$h^{\alpha\beta} T_{\alpha\beta} = 0$$

as claimed.

Since there is ambiguity in the choice of h , our string theory has redundant D.O.F.

$\rightarrow$ We can fix a gauge for the theory.

Claim: We can fix the Gauge of h in such a way that h is Conformal

to the flat metric in M^2 . $\exists \phi$ s.t.
i.e. $h_{\alpha\beta} = e^{2\phi(\sigma)} \eta_{\alpha\beta}^{(2)}$

Proof: Since $h_{\alpha\beta} \in \text{Sym}^2(\mathbb{R})$

$h_{10} = h_{01} \Leftrightarrow \exists$ only 3 ind. param.

- Diffeomorphism invariance $f\sigma \mapsto \tilde{\sigma}$ allows us to fix two of the components $h_{10} = 0 = h_{01}$ and $h_{00} = \pm h_{11}$

$\Rightarrow h_{\alpha\beta} = h(x) \eta_{\alpha\beta}$

- From Weyl invariance we can remove $h(x)$ to obtain $\eta_{\alpha\beta}$!

$\Rightarrow$ Gauged: $\boxed{h_{\alpha\beta} = \eta_{\alpha\beta}^{(2)}}$

- This is only true locally if the Bundle/
Mfd are trivial & have no topological
obstructions.

* Geometrically: This means that the world-sheet Σ must have vanishing Euler-characteristic $\chi(\Sigma)$ as

$$0 = \int d^2\sigma \sqrt{-h} R^{(2)} = \chi(\Sigma)$$

Flat World Sheet String Action:

$$S = T \int d^2\sigma \sqrt{-h} \eta^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu$$

$$S = \frac{1}{2} \int d^2\sigma \eta^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\mu$$

$$= -\frac{T}{2} \int d^2\sigma (\dot{X}^2 - \vec{X}'^2)$$

where $X^\mu = dx^\mu/d\sigma$, $\dot{X}^\mu = \frac{dX^\mu}{d\tau}$

* These are the local conditions. However we can do better. Let us consider the global Criteria

$$\rightarrow h_{\alpha\beta} = e^{-2\phi} \tilde{h}_{\alpha\beta}$$

$$\Leftrightarrow \delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha - 2\lambda_0 h_{\alpha\beta}$$

$$=: (P\xi)_{\alpha\beta} - 2\tilde{\lambda}_0 h_{\alpha\beta}$$

where $P: TM \rightarrow \text{Sym}^2(TM)$ i.e.

P takes a vector to a symmetric-traceless rank 2 tensor.

$$(P\xi)_{\alpha\beta} := \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha - (\nabla_\gamma \xi^\gamma) h_{\alpha\beta}$$

and

$$2\tilde{\lambda}_0 := -2\lambda_0 + \nabla_\gamma \xi^\gamma$$

Now, note that as with Killing vectors we found that $x^\mu \mapsto x^\mu + \xi^\mu(x) \Leftrightarrow \xi^\mu \in TM \Leftrightarrow \delta x^\mu = \mathcal{L}_\xi x^\mu$

Here $\delta h_{\alpha\beta} \in TM_g$, so this variation is a tangent vector in the space of metrics $\mathcal{M}_g$.

→ See Nakamura for a geometric view of the Teichmüller space

$$\mathcal{T}_g := \mathcal{M}_g / (\text{diff}_0(\Sigma_g) \rtimes \text{Weyl}(\Sigma_g))$$

on this tangent space we can define an inner product:

$$(\delta h^{(1)}, \delta h^{(2)}) := \int d^2\sigma \sqrt{-h} h^{\alpha\gamma} h^{\beta\delta} \delta h_{\alpha\beta}^{(1)} \delta h_{\gamma\delta}^{(2)}$$

With this, we find that $(P\xi)_{\alpha\beta}$ is orthogonal to the trace of $\delta h_{\alpha\beta}$:

$$h^{\alpha\beta} \delta h_{\alpha\beta} = 2 \nabla_\gamma h^\gamma = 2 \nabla = 2 (\underbrace{\nabla_\gamma h^\gamma}_{\text{trace of } h_{\alpha\beta}} - 2\lambda_D)$$

thus, without loss of generality we can choose $\lambda_D = \frac{1}{2} \nabla_\gamma h^\gamma$, which removes the trace part SEEMINGLY leaving the variation $\delta h_{\alpha\beta}$ invariant since (replacing $\nabla_\gamma h^\gamma = 2\lambda_D$)

$$(P\xi)_{\alpha\beta} = 2 (\nabla_\alpha \xi_{\beta} - \delta_{\alpha\beta} \lambda_D)$$

* We conclude, $h_{\alpha\beta} = e^{-2\phi} \eta_{\alpha\beta}$ implies $\exists$ a globally defined vector field ξ^α such that

$$(P\xi)_{\alpha\beta} = \mathcal{L}_{\xi} h_{\alpha\beta}$$

for an arbitrary symmetric traceless $t_{\alpha\beta}$.

* Note, if $\exists$ zero modes of P i.e.

ξ_0 s.t. $(P\xi_0) = 0$ then $\forall$ solutions ξ , $\xi + \xi_0$ is also a solution.

• The adjoint of P ; $P^T: \text{Sym}^2(T\Sigma) \rightarrow T\Sigma$

$$(P^T t)_\alpha := -2\nabla^\beta t_{\alpha\beta}$$

The zero modes of P^T are elements of $\text{Sym}^2(T\Sigma)$ s.t. $t_{\alpha\beta}$ cannot be written as $(P\xi)_{\alpha\beta} \forall \xi$.

Proof: If t_0 is s.t. $(P^T t_0)_\alpha = 0$

then $\forall \xi : (\xi, P^T t_0) = 0$

$$\Rightarrow (P\xi, t_0) = 0 \Rightarrow P\xi = 0$$

From P we have

$$T_g := P T_g \oplus P^\perp T_g$$

$$P T_g := \{t_{\alpha\beta} \sim (P\xi)_{\alpha\beta}\}$$

$$(P\xi) = 0 \text{ (Conformal Killing eq)}$$

Field Equations for Sp :

* Let us fix our gauge $\hookrightarrow h_{\alpha\beta} = \eta_{\alpha\beta}$

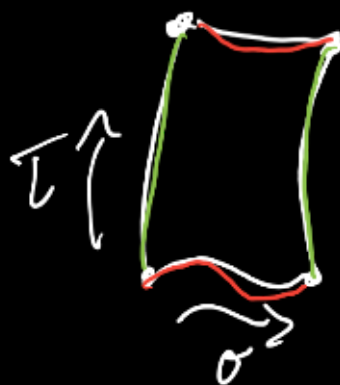
Let $X^\mu(\tau, \sigma) \mapsto X^\mu(\tau, \sigma) + \delta X^\mu(\tau, \sigma)$

$$\therefore \delta S_p = \delta \left(\frac{T}{2} \int d\tau d\sigma (\dot{X}^2 - (X')^2) \right)$$

$$= T \int d\tau d\sigma (\dot{X}^\mu \delta \dot{X}_\mu - X'^\mu \delta X'_\mu)$$

$$= T \int d\tau d\sigma \left(\dot{X}^\mu \frac{d}{d\tau} (\delta X_\mu) - X'^\mu \frac{d}{d\sigma} (\delta X_\mu) \right)$$

$$\begin{aligned}
 &= T \int d\tau d\sigma \left[\frac{d}{d\tau} (\dot{X}^\mu \delta X_\mu) - \ddot{X}^\mu \delta X_\mu \right. \\
 &\quad \left. - \frac{d}{d\sigma} (X'^\mu \delta X_\mu) + X''^\mu \delta X_\mu \right] \\
 &= T \int d\sigma (\dot{X}_\mu \delta X^\mu) \Big|_{\partial\tau} - T \int d\tau (X_\mu' \delta X^\mu) \Big|_{\sigma=0}^{\sigma=\pi} \\
 &\quad + T \int d\sigma d\tau (X_\mu'' - \ddot{X}_\mu) \delta X^\mu
 \end{aligned}$$



$\begin{pmatrix} \partial\tau \\ \partial\sigma \end{pmatrix}$ It's clear we must Not have a tangent vector at the boundary of $\partial\tau$

• For the classical particle we have



and $\delta x_1 = \delta x_2 = 0$

$\forall t$, meaning the worldsheet is fixed for that time.

Wouldn't make sense to allow for varying end points when maximizing Sp.

$$\therefore \boxed{\int d\sigma (\dot{X}_\mu \delta X^\mu) \Big|_{\partial\tau} = 0}$$

$$\Rightarrow \delta S = 0$$

$$\neq 0$$

$$\boxed{(-\partial_\tau^2 + \partial_\sigma^2) X^\mu - T \int d\tau \left[X_\mu' \delta X^\mu \Big|_{\sigma=\pi} - X_\mu' \delta X^\mu \Big|_{\sigma=0} \right]}$$

* This Boundary

Boundary term (B)

term determines the type of string we have.

(1) Closed String:

For a closed string we take $0 \sim \sigma + l$

$$X^\mu(\tau, \sigma + l) = X^\mu(\tau, \sigma)$$

This trivially satisfies (B) since

$$\delta X^\mu(\tau, \sigma + l) = \delta X^\mu(\tau, \sigma) \Rightarrow (B) = 0.$$

This leaves

$$(\partial_\tau^2 + \partial_\sigma^2) X^\mu(\tau, \sigma) = 0$$

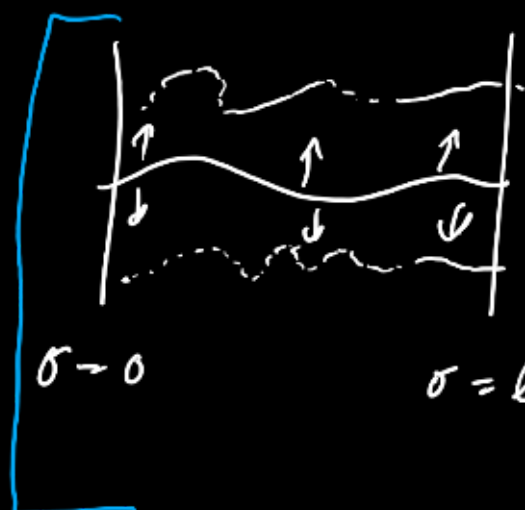
$$\square X^\mu(\tau, \sigma) = 0$$

$$\text{B.C.'s: } X^\mu(\tau, \sigma + l) = X^\mu(\tau, \sigma)$$

Open strings: There are two possibilities

$$\partial_\sigma X^\mu|_{\sigma=0,l} = 0 \quad (\text{Neumann B.C.'s})$$

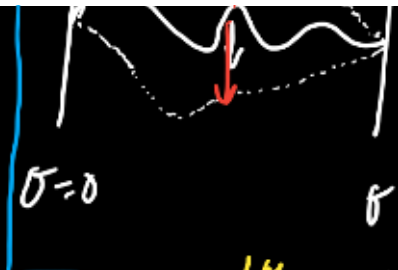
$$\delta X^\mu|_{\sigma=0,l} = 0 \quad (\text{Dirichlet B.C.'s})$$



Neumann: The string can oscillate and its endpoints can move along $\partial\sigma$ as long as the vectors $\partial_\sigma X^\mu = 0$ at $\partial\sigma$



Dirichlet: String can oscillate but its endpoints are fixed



but its end points are fixed

$$\delta X^M|_{\sigma=0,l} = 0$$

$$x'^M - x^M = \delta X^M$$

This is why $\delta X^M|_{\partial\sigma} = 0$ for Dirich since $\uparrow = 0$ at $\partial\sigma$.

Poincare:

- Neumann: These B.C.'s $\partial_\sigma X^M|_{\partial\sigma} = 0$ are preserved under Poincaré

$$\begin{aligned}\partial_\sigma (X'^M)|_{\sigma=0,l} &= \partial_\sigma (\lambda^M_N X'^N + a^M)|_{\sigma=0,l} \\ &= \lambda^M_N \partial_\sigma X'^N|_{\sigma=0,l} = 0 \quad \checkmark\end{aligned}$$

- Dirichlet:

Let us be slightly more concrete.

Since $\delta X^M(\tau, 0) = \delta X^M(\tau, l) = 0$

it's clear that $X^M|_{\partial\sigma}$ is

constant. So we define the following

$$X^M(\tau, 0) =: X^M_0$$

$$X^M(\tau, l) =: X^M_l$$

Although $X^M_0 = X^M_0(\tau)$ it's clear that this does not vary in time either so this really is a constant.

Claim: D. B.C.'s do not preserve the IMP

Poincare symmetry:

Proof: $X'^{\mu}|_{\sigma=0,e} = (\lambda^{\mu}_{\nu} X'^{\nu} + b^{\mu})|_{\sigma=0,e}$
 $= \lambda^{\mu}_{\nu} (X'^{\nu})|_{\sigma=0,e} + b^{\mu}$
 $\neq X'^{\mu}_{0,m}$

* Dirichlet boundary conditions give rise to D_p -branes.

Suppose we have Neumann B.C.'s on $p+1$ background spacetime coord's.

$$\partial_{\sigma} X^{\mu}|_{\sigma=0,e} = 0, \quad \mu = 1, \dots, p+1$$

and Dirichlet B.C.'s on $D-(p+1)$

coords $X^{\nu}_{0,e} \equiv \nu = p+1, \dots, D$

$\Rightarrow$ The string ends on a D_p -brane.

Solving the field Eqs: $(\partial_{\sigma}^2 - \partial_{\tau}^2) X^{\mu} = 0$

• Because we have gauge fixed $\eta_{\alpha\beta} = \eta_{\alpha\beta}$ we must impose the tracelessness of the E.M. tensor

$$\eta^{\alpha\beta} T_{\alpha\beta} = 0 \Leftrightarrow \partial_{\alpha} X^{\mu} \partial_{\beta} X_{\mu} = \frac{1}{2} \eta_{\alpha\beta} \eta^{\delta\delta} \partial_{\delta} X^{\mu} \partial_{\delta} X_{\mu}$$

or $T_{00} = T_{11} = \frac{1}{2} (\dot{X}^2 + X'^2) = 0$

$$T_{01} = T_{10} = \dot{X}_{\mu} X'^{\mu} = 0$$

light-cone coordinates: $\sigma^\pm := \tau \pm \sigma$

$$\sigma = \frac{1}{2}(\sigma^+ - \sigma^-), \quad \tau = \frac{1}{2}(\sigma^+ + \sigma^-)$$

$$\partial_t = \partial/\partial\sigma^+ = \frac{\partial\tau}{\partial\sigma^+} \frac{\partial}{\partial\tau} + \frac{\partial\sigma}{\partial\sigma^+} \frac{\partial}{\partial\sigma}$$

$$\Rightarrow \frac{1}{2}(\partial_\tau + \partial_\sigma) = \partial_+$$

$$\partial_- = \frac{1}{2}(\partial_\tau - \partial_\sigma)$$

$$\eta_{\pm\pm} = -\frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\eta_{+-} = -1/2, \quad \eta_{++} = 0$$

$$\eta_{-+} = -1/2, \quad \eta_{--} = 0$$

$$\eta^{\pm\pm} = -2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$\therefore$ The E.O.M becomes

$$(-\partial_\sigma^2 + \partial_\tau^2)X^\mu = 0$$

$$(\partial_\tau - \partial_\sigma)(\partial_\tau + \partial_\sigma)X^\mu = 0$$

$$\partial_- \partial_+ X^\mu = 0 \quad (a)$$

$$\Rightarrow T_{++} = \partial_+ X^\mu \partial_+ X_\mu = 0 \quad (b)$$

$$T_{--} = \partial_- X^\mu \partial_- X_\mu = 0 \quad (c)$$

a, b, c are the equations we must solve for.

(Not in many textbooks)

Proof of mode expansion for Closed Strings:

We start in $1D^2 \cong \Sigma$ and $(\partial_\sigma^2 - \partial_\tau^2)X^\mu = 0$

(i) Assume $X^\mu(\tau, \sigma) = g(\tau) f^\mu(\sigma)$

$$\Leftrightarrow \left(\frac{d^2 f^\mu(\sigma)}{d\sigma^2} \right) - \frac{f^\mu(\sigma)}{g(\tau)} \frac{d^2 g(\tau)}{d\tau^2} = 0$$

$$\Leftrightarrow \exists c \in \mathbb{R} \quad \text{s.t.} \quad \frac{d^2 f^\mu(\sigma)}{d\sigma^2} = c f^\mu(\sigma) \quad \text{and} \quad \frac{d^2 g(\tau)}{d\tau^2} = c g(\tau)$$

We assume that $c = -\alpha'^2$

$$\frac{d^2 f^\mu}{d\sigma^2} = -\lambda^2 f^\mu(\sigma) \quad \therefore \dots$$

$$f^\mu(\sigma) = A^\mu \cos(\sigma\lambda) + B^\mu \sin(\sigma\lambda)$$

$$g(\tau) = A \cos(\tau\lambda) + B \sin(\tau\lambda)$$

We must now impose Closed String B.C.'s

$$\rightarrow f^\mu(0) = f^\mu(l) = A^\mu \cos(l\lambda) + B^\mu \sin(l\lambda) \\ = A^\mu$$

$$\Rightarrow \sin(l\lambda) = 0 \Leftrightarrow l\lambda = n\pi, \quad n \in \mathbb{Z} \\ \lambda = \frac{n\pi}{l}$$

However we must also have

$$\cos(l\lambda) = 1 \Rightarrow l\lambda = 2\pi n$$

$$\boxed{\lambda = 2\pi n/l}$$

$$\text{Likewise } g(\tau) = A \cos(2\pi n\tau/l) + B \sin(2\pi n\tau/l)$$

$\rightarrow$ Go to $e^{\pm i}$:

$$f^\mu(\sigma) = \tilde{A}^\mu e^{-i\frac{2\pi n}{l}\sigma} + \tilde{B}^\mu e^{i\frac{2\pi n}{l}\sigma}$$

$$g(\tau) = \tilde{A} e^{-i\frac{2\pi n}{l}\tau} + \tilde{B} e^{i\frac{2\pi n}{l}\tau}$$

$$\Leftrightarrow \boxed{\begin{aligned} (f^\mu g_\mu) = x^\mu_{(\lambda)} &= \tilde{A}^\mu \tilde{A} e^{-i\lambda(\sigma+\tau)} + \tilde{B}^\mu \tilde{B} e^{i\lambda(\sigma+\tau)} \\ &+ \tilde{B}^\mu \tilde{A} e^{-i\lambda(\tau-\sigma)} + \tilde{A}^\mu \tilde{B} e^{i\lambda(\tau-\sigma)} \end{aligned}}$$

check

$$x^\mu(0,0) = x^\mu(\tau,l)$$

$$\tilde{A}^\mu e^{i\lambda\tau} + B^\mu e^{i\lambda\tau} + C^\mu e^{-i\lambda\tau} + D^\mu e^{-i\lambda\tau}$$

$$= e^{-i\lambda\tau} (A^\mu + C^\mu) + e^{i\lambda\tau} (B^\mu + D^\mu)$$

$\rightarrow$ Consistency:

$$e^{-i\lambda(\tau+\sigma)} = e^{-i\lambda(\tau-\sigma)}$$

$$f^{\mu}(\sigma)g(\tau) = A^{\mu} e^{-in(\tau+\sigma)} + B^{\mu} e^{-in(\tau-\sigma)}$$

$$n \neq 0$$

$$\text{For } n=0 \Rightarrow \lambda=0$$

$$\frac{\partial^2 f^{\mu}}{\partial \sigma^2} = 0 \Rightarrow a^{\mu} \sigma + b^{\mu} = f_{(0)}^{\mu}(\sigma)$$

$$f_{(0)}^{\mu}(\ell) = a^{\mu} \ell + b^{\mu} = b^{\mu} \Rightarrow a^{\mu} = 0$$

$$\therefore f_{(0)}^{\mu}(\sigma) = 0$$

$$g(\tau) = a\tau + b \quad (\text{no restrictions})$$

$$\begin{aligned} X^{\mu}(\sigma, \tau) &= b^{\mu}(a\tau + b) + \sum_{n \neq 0} A_n^{\mu} e^{-in(\tau+\sigma)} + \sum_{n \neq 0} B_n^{\mu} e^{-in(\tau-\sigma)} \\ &= b^{\mu}b + ab^{\mu}\tau + \sum_{n \neq 0} \left(\right) \end{aligned}$$

$$\therefore b^{\mu}b \equiv: x^{\mu}$$

$$ab^{\mu} \equiv: l_s^2 p^{\mu}$$

$$X^{\mu}(\sigma, \tau) = x^{\mu} + \tau l_s^2 p^{\mu} + \sum_{n \neq 0} A_n^{\mu} e^{-in(\tau+\sigma)} + B_n^{\mu} e^{-in(\tau-\sigma)}$$

Use last conditions

$$\dot{X}^{\mu} = l_s^2 p^{\mu} + \sum_{n \neq 0} A_n^{\mu} (-in) e^{-in(\tau+\sigma)} + (-in) B_n^{\mu} e^{-in(\tau-\sigma)}$$

$$X'^{\mu} = \sum_{n \neq 0} A_n^{\mu} (-in) e^{-in(\tau+\sigma)} + B_n^{\mu} (in) e^{-in(\tau-\sigma)}$$

$$X^{\mu} = X_R^{\mu} + X_L^{\mu} = \underbrace{x^{\mu} + \tau l_s^2 p^{\mu}}_{\text{center of mass}} + \underbrace{\frac{i}{2} l_s^2 \sum_{n \neq 0} \frac{1}{n} (\alpha_n^{\mu} e^{2in\sigma} + \tilde{\alpha}_n^{\mu} e^{-2in\sigma})}_{\text{oscillators}} e^{-2in\tau}$$

$$T = \frac{1}{2\pi\alpha'}, \quad \alpha' = \frac{1}{2}l_s^2$$

$$X_R^\mu = \frac{1}{2}x^\mu + \frac{1}{2}l_s^2(\tau - \sigma)p^\mu + \frac{i}{2}l_s \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in(\tau - \sigma)}$$

$$X_L^\mu = \frac{1}{2}x^\mu + \frac{1}{2}l_s^2(\tau + \sigma)p^\mu + \frac{i}{2}l_s \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_n^\mu e^{-in(\tau + \sigma)}$$

- Note the first term looks like the trajectory of a point particle

$$\frac{d^2}{d\tau^2}(X^\mu) = 0 \Rightarrow X^\mu = x^\mu + p^\mu \tau$$

$\Rightarrow X^\mu$ must be real

$$\Rightarrow x^\mu, p^\mu \in \mathbb{R}$$

$$\text{Note: } \left(\sum_{n \neq 0} \frac{1}{n} (\alpha_n^\mu)^* e^{in(\tau - \sigma)} \right)$$

let $n \mapsto -n$, then

$$\sum_{n \neq 0} \frac{1}{(-n)} (\alpha_{-n}^\mu)^* e^{-in(\tau - \sigma)}$$

$$\Leftrightarrow \boxed{(\alpha_{-n}^\mu)^* = \alpha_n^\mu} \quad \boxed{\tilde{\alpha}_{-n}^\mu = (\tilde{\alpha}_n^\mu)^*}$$

- Recall $\Pi^\mu(\tau, \sigma) = \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} = T \dot{X}^\mu = \frac{\dot{X}^\mu}{\pi \alpha'}$
 $= \frac{p^\mu}{\pi} + \frac{1}{\pi l_s} \sum_{n \neq 0} \left(\alpha_n^\mu e^{-in(\tau - \sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau + \sigma)} \right)$

* We find this momenta satisfy the following

Poisson bracket:

$$\{ \Pi^\mu(\tau, \sigma), \Pi^\nu(\tau, \sigma') \}_{\text{P.B.}} = 0$$

$$\{ X^\mu(\tau, \sigma), X^\nu(\tau, \sigma') \}_{\text{P.B.}} = 0$$

$$\{ \Pi^\mu(\tau, \sigma), X^\nu(\tau, \sigma') \}_{\text{P.B.}} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

$$\begin{aligned} \{ \alpha_m^\mu, \alpha_n^\nu \}_{P.B.} &= \{ \tilde{\alpha}_m^\mu, \tilde{\alpha}_n^\nu \}_{P.B.} = i m n \eta^{\mu\nu} \delta_{m,-n} \\ \{ \alpha_m^\mu, \tilde{\alpha}_n^\nu \}_{P.B.} &= 0 \\ \{ p^\mu, x^\nu \}_{P.B.} &= \eta^{\mu\nu} \end{aligned}$$

Proof: $p^\mu(\tau, \sigma) = \frac{p^\mu}{\pi} + \frac{1}{\pi l_s} \sum_{n \neq 0} (\alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)})$

$$p^\mu(\tau, \sigma) p^\nu(\tau, \sigma') = \left(\frac{p^\mu p^\nu}{\pi^2} + p^\mu \frac{1}{\pi l_s} \sum_{n \neq 0} \left(\right) + p^\nu \frac{1}{\pi l_s} \sum_{n \neq 0} \left(\right) + \left(\frac{1}{\pi l_s} \right)^2 \sum_{n \neq 0} \left(\right) \sum_{m \neq 0} \left(\right) \right)$$

• For simplicity: $\sigma' = 0$: Project the eigen basis of $\frac{\partial^2}{\partial \sigma^2}$ (i.e. do

a Fourier transform:

$$p^\mu(\tau, \sigma) p^\nu(\tau, 0) = \frac{p^\mu p^\nu}{\pi^2} + \frac{p^\mu}{\pi l_s} \sum_{n \neq 0} (\alpha_n^\nu e^{-2in\tau} + \tilde{\alpha}_n^\nu e^{-2in\tau}) + \frac{p^\nu}{\pi l_s} \sum_{n \neq 0} (\alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)}) + \left(\frac{1}{\pi l_s} \right)^2 \sum_{n \neq 0} \sum_{m \neq 0} (\alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)}) (\alpha_m^\nu e^{-2im\tau} + \tilde{\alpha}_m^\nu e^{-2im\tau})$$

$$+ m \rightarrow n \quad \stackrel{!}{=} 0$$

$$\bullet \sum_n \sum_m \left[\alpha_n^\mu \alpha_m^\nu e^{-2in(2\tau-\sigma)} + \tilde{\alpha}_n^\mu \alpha_m^\nu e^{-2in(2\tau+\sigma)} + \alpha_n^\mu \tilde{\alpha}_m^\nu e^{-2in(2\tau-\sigma)} + \tilde{\alpha}_n^\mu \tilde{\alpha}_m^\nu e^{-2in(2\tau+\sigma)} \right]$$

(ii) $\{ X^\mu(\tau, \sigma), X^\nu(\tau, 0) \}_{P.B.} \stackrel{!}{=} 0$

ΣΣ term:

$$-\frac{1}{2} l_s^2 \sum_n \sum_m (\alpha_n^\mu e^{2in\sigma} + \tilde{\alpha}_n^\mu e^{-2in\sigma}) (\alpha_m^\nu + \tilde{\alpha}_m^\nu) e^{-4i(n+m)\tau}$$

$$= -\frac{1}{4} l_s^2 \sum_n \sum_m (\alpha_n^\mu \alpha_m^\nu e^{2in\sigma} + \tilde{\alpha}_n^\mu \alpha_m^\nu e^{-2in\sigma} + \alpha_n^\mu \tilde{\alpha}_m^\nu e^{2in\sigma} + \tilde{\alpha}_n^\mu \tilde{\alpha}_m^\nu e^{-2in\sigma}) e^{-4i(n+m)\tau}$$

→ Much simpler to apply $\{ \cdot, \cdot \}$:

$$\begin{aligned} \{ p^\mu(\tau, \sigma), p^\nu(\tau, 0) \}_{P.B.} &= 0 \Leftrightarrow \\ \left\{ p^\mu + \frac{1}{\pi l_s} \sum_n (\alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)}), \frac{p^\nu}{\pi} + \frac{1}{\pi l_s} \sum_n (\alpha_n^\nu e^{-2in\tau} + \tilde{\alpha}_n^\nu e^{-2in\tau}) \right\} \\ &= \frac{1}{\pi} \{ p^\mu, p^\nu \} + \frac{1}{\pi l_s^2} \sum_n \left\{ p^\mu, \alpha_n^\nu \right\} e^{-2in\tau} + \{ p^\mu, \tilde{\alpha}_n^\nu \} e^{-2in\tau} \\ &\quad + \frac{1}{\pi l_s^2} \sum_n \left\{ \alpha_n^\mu, p^\nu \right\} e^{-2in(\tau-\sigma)} + \{ \tilde{\alpha}_n^\mu, p^\nu \} e^{-2in(\tau+\sigma)} \\ &\quad + \frac{1}{(\pi l_s)^2} \sum_n \sum_m \left(\{ \alpha_n^\mu, \alpha_m^\nu \} e^{-2in(2\tau-\sigma)} + \{ \alpha_n^\mu, \tilde{\alpha}_m^\nu \} e^{-2in(2\tau+\sigma)} + \{ \tilde{\alpha}_n^\mu, \alpha_m^\nu \} e^{-2in(2\tau-\sigma)} \right. \\ &\quad \left. + \{ \tilde{\alpha}_n^\mu, \tilde{\alpha}_m^\nu \} e^{-2in(2\tau+\sigma)} \right) \end{aligned}$$

for each term $\alpha_n^\mu \in \mathbb{C}$.

$$\begin{aligned} \{p^\mu, p^\nu\} &= 0 \\ \{\alpha_n^\mu, p^\nu\} e^{-2in\tau} + \{\tilde{\alpha}_n^\mu, p^\nu\} e^{2in\tau} &= 0 \\ \{p^\mu, \alpha_n^\nu\} e^{-2in\tau} + \{p^\mu, \tilde{\alpha}_n^\nu\} e^{2in\tau} &= 0 \\ \{\alpha_n^\mu, \alpha_n^\nu\} e^{-2i(n+m)\tau} + \dots + \dots &= 0 \end{aligned}$$

→ Simply equate coefficients of terms $e^{im\sigma}$ ($m \neq 0$) etc..

- $f_n(\sigma) := e^{in\sigma}/\sqrt{2\pi}$, $n \in \mathbb{Z}$ form a complete basis for f^{ns} defined for $\sigma \in [0, 2\pi]$. Any $f^\mu \sim g(\sigma, \sigma') \sim \sum_{n,m} f_n(\sigma) f_m(\sigma')$
- Similarly for $\{x^\mu(\tau, \sigma), x^\nu(\tau, \sigma)\} = 0$ & $\{p^\mu(\tau, \sigma), x^\nu(\tau, \sigma)\} = \eta^{\mu\nu}$.

$$\therefore \{p^\mu, p^\nu\} = \{x^\mu, x^\nu\} = 0, \quad \{p^\mu, x^\nu\} = \eta^{\mu\nu}$$

$$\{\alpha_n^\mu, \alpha_n^\nu\} = \{\tilde{\alpha}_n^\mu, \tilde{\alpha}_n^\nu\} = i n \eta^{\mu\nu} \delta_{n+m, 0}, \quad \{\alpha_n^\mu, \tilde{\alpha}_n^\nu\} = 0$$

* The classical Poisson bracket is defined by a symplectic structure of the embedded M^{fd}

$$\{F, G\} := \int d\sigma \left(\frac{\partial F(\tau, \sigma)}{\partial x^\mu(\tau, \sigma)} \frac{\partial G(\tau, \sigma')}{\partial \pi_\mu(\tau, \sigma')} - \frac{\partial G(\tau, \sigma')}{\partial x^\mu(\tau, \sigma)} \frac{\partial F(\tau, \sigma)}{\partial \pi_\mu(\tau, \sigma)} \right)$$

The Hamiltonian is given by $H = \int_0^l d\sigma \underbrace{(\dot{x}^\mu(\tau, \sigma) \pi_\mu(\tau, \sigma) - \mathcal{L})}_{\text{Hamiltonian Density} \rightarrow \mathcal{H}}$

Open strings:

(i) Neumann B.C's. $\partial_\sigma X^\mu(\tau, \sigma)|_{\sigma=0, l} = 0$

→ These give a much different solution space

$$X^\mu(\tau, \sigma) = f(\sigma) g(\tau) \Leftrightarrow \frac{\partial^2 f(\sigma)}{\partial \sigma^2} \frac{1}{f} = \lambda$$

$$f(\sigma) = e^{\pm i\sqrt{\lambda} \sigma}, \quad g(\tau) = e$$

$$\partial_\sigma f(\sigma) = \pm i\sqrt{\lambda} e^{\pm i\sqrt{\lambda} \sigma} \Big|_{0, l} \stackrel{!}{=} 0$$

$\Rightarrow \lambda = 0$

$$f^{\mu}(\sigma) = a^{\mu} \sigma + b^{\mu}$$

$$|_{\sigma=l} = \partial_{\sigma} f^{\mu}(\sigma) = A^{\mu} i\sqrt{\lambda} e^{i\sqrt{\lambda} \sigma} + B^{\mu} (-i\sqrt{\lambda}) e^{-i\sqrt{\lambda} \sigma}$$

$$\hookrightarrow A^{\mu} e^{i\sqrt{\lambda} l} + B^{\mu} e^{-i\sqrt{\lambda} l} = 0$$

$$B = A^{\mu} e^{2i\sqrt{\lambda} l}$$

Assuming $\frac{\partial^2 f}{\partial \sigma^2} = -\lambda^2$

$$f^{\mu}(\sigma) = A^{\mu} \cos(\lambda \sigma) + B^{\mu} \sin(\lambda \sigma)$$

$$\partial_{\sigma} f^{\mu}(\sigma) = -\lambda A^{\mu} \sin(\lambda \sigma) + \lambda B^{\mu} \cos(\lambda \sigma)$$

$$|_{\sigma=0} : \lambda B^{\mu} = 0$$

$\bullet \lambda = 0 \Rightarrow \underline{f^{\mu}(\sigma) = a^{\mu} \sigma + b^{\mu}}$

$\bullet B^{\mu} = 0 \quad \underline{f^{\mu}(\sigma) = A^{\mu} \cos(\lambda \sigma)}$

For $\partial_{\sigma} X^{\mu}(\tau, \sigma)|_{\sigma=l} = 0$

$$\hookrightarrow \underline{f^{\mu}(\sigma) = a^{\mu}}$$

$$\hookrightarrow \sin(\lambda \sigma) = 0 \Leftrightarrow \lambda l = \pi n, \quad n \in (\mathbb{Z} \setminus \{0\})$$

$$\underline{f^{\mu}(\sigma) = A^{\mu} \cos\left(\frac{\pi n}{l} \sigma\right)}$$

w/ No constraints on $g(\tau) = A\tau + b$

$$g_{\mu\nu}(\tau) = (A e^{-i\pi\tau} + B e^{i\pi\tau})$$

$$X^{\mu}(\tau, \sigma) = a^{\mu} (A\tau + b) \quad (\text{See 4.2 for Dirichlet}) \dots$$

Symmetries in Canonical Quantization:

• For a global symmetry of the system on the worldsheet

$$\exists j^a : \partial_a j^a = 0 \Leftrightarrow \frac{d}{dt} Q = \frac{d}{dt} \left(\int d\sigma j^0 \right) = 0$$

$$\frac{d}{dt} \left(\int d\sigma j^0 \right) = \int d\sigma \frac{d}{dt} j^0$$

however $\partial_a j^a = \partial_t j^0 + \partial_{\sigma} j^{\sigma} = 0$

$$\frac{dQ}{dt} = - \int_0^l d\sigma \partial_{\sigma} j^{\sigma} = - j^{\sigma} \Big|_{\sigma=0}^{\sigma=l} = 0$$

For $U: \phi \mapsto \phi + \epsilon \phi$, $\delta \phi = \epsilon U(\phi)$

$\leadsto$ Take ϵ to be local then

$$\delta S = \int d\tau d\sigma (\partial_a \epsilon) j^a = - \int d\tau d\sigma \epsilon \partial_a j^a$$

(i) Poincaré' Charge/Current for Bosonic string:

(a) Translation $x^\mu \mapsto x^\mu + a^\mu(\sigma)$

$$S_p = -\frac{T}{2} \int d\tau d\sigma \eta^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x_\mu$$

$$\hookrightarrow \delta S_p = -T \int d\tau d\sigma \partial_\alpha (a^\mu(\tau)) \partial^\alpha x_\mu$$

$$\Rightarrow \boxed{j^\alpha_\mu = -T \partial^\alpha x_\mu}$$

This implies the current can be written as

$$j^\alpha_\mu = -T \partial^\alpha \eta_{\mu\nu} x^\nu \hookrightarrow \underline{j^{\alpha\mu} = -T \partial^\alpha x^\mu}$$

* Interpret this as $\exists \mu$ currents j^α for each $g^{\mu\nu} x^\mu$ which embed the string.

To see that this current is conserved:

$$\partial_\alpha j^{\alpha\mu} = -T \partial_\alpha \partial^\alpha x^\mu = -T (-\partial_\tau^2 + \partial_\sigma^2) = 0$$

On-shell

• The conserved charge $Q = \int d\sigma j^0$ is naturally going to be the string momentum

$$p^\nu = \int d\sigma (-T \partial^0 x^\nu) = T \int d\sigma \partial_\tau x^\nu = T \int d\sigma \dot{x}^\nu = \int d\sigma T P^\nu$$

P^ν : Canonical momentum, conjugate to $x^\mu \Rightarrow P^\nu = \pi^\nu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\nu}$

Thus, we see that p^μ in the $x^\mu(\tau, \sigma)$ was chosen to show that the action S_σ has translational symmetry.

(b) Lorentz transformations: $x^\mu \mapsto (\Lambda^{-1})^\mu_\nu x^\nu$

$$\Rightarrow (\Lambda^{-1})^\mu_\nu = (\Lambda^{-1})^\mu_\nu(x)$$

$$\delta S_p = -T \int d\sigma d\tau \partial_\alpha (\Lambda^{-1})^\mu_\nu x^\nu \partial^\alpha x_\mu$$

$$= -T \int d\sigma d\tau \left[\partial_\alpha (\Lambda^{-1})^\mu_\nu x^\nu \partial^\alpha x_\mu + (\Lambda^{-1})^\mu_\nu (\partial_\alpha x^\nu) \partial^\alpha x_\mu \right]$$

The second term is:

$$\underbrace{(\Lambda^{-1})^\mu_\nu}_{\text{Anti}} \underbrace{\partial_\alpha x^\nu \partial^\alpha x_\mu}_{\text{Sym}} \stackrel{?}{=} 0$$

$$\Rightarrow \delta S_p = -T \int d\sigma d\tau \partial_\alpha (\Lambda^{-1})^\mu_\nu X^\nu \partial^\alpha X_\mu = \int d\sigma d\tau (\partial^\alpha \Lambda_{\mu\nu}) X^\mu \partial_\alpha X^\nu$$

$$\therefore j_\alpha^{\mu\nu} = -\frac{T}{2} (X^\mu \partial_\alpha X^\nu - X^\nu \partial_\alpha X^\mu)$$

Q: How do we know to "make" $j_\alpha^{\mu\nu}$ antisymmetric? Note that $\Lambda_{\mu\nu}$ is antisymmetric and hence so too should the current.

- $j_\alpha^{\mu\nu}$ is only conserved on-shell.

Hamiltonian & E.M. Tensor:

- Hamiltonian density $\mathcal{H} = \dot{X}_\mu p^\mu - \mathcal{L}$

$$\text{Hamiltonian: } H := \int_0^l d\sigma (\mathcal{H})$$

$$H = \int_0^l d\sigma (\dot{X}^\mu p_\mu - \mathcal{L})$$

For the Bosonic string $p^\mu = T \dot{X}^\mu$ and $\Sigma \cong M^2: \mathcal{L} = \frac{1}{2} G_{\mu\nu} \dot{X}^\mu \dot{X}^\nu$.

$$S = \int d\tau d\sigma \left(-\frac{T}{2} \right) (-\partial_\tau X^\mu \partial_\tau X_\mu + \partial_\sigma X^\mu \partial_\sigma X_\mu)$$

$$H = \int d\sigma \left(T \dot{X}^2 - \frac{T}{2} \dot{X}^2 + \frac{T}{2} X'^2 \right) = \frac{T}{2} \int d\sigma (\dot{X}^2 + X'^2)$$

$$\therefore H = \frac{T}{2} \int d\sigma (\dot{X}^2 + X'^2) = T \int_0^l d\sigma (\partial_+ X)^2 + (\partial_- X)^2$$

- To understand how to interpret H we must expand X_μ^R its oscillation modes.

Closed string Hamiltonian:

$$\dot{X}^\mu = \frac{1}{l_s} p^\mu + \frac{1}{l_s} \sum_{n \neq 0} (\alpha_n^\mu e^{2in\sigma} + \tilde{\alpha}_n^\mu e^{-2in\sigma}) e^{-2in\tau}$$

$$\therefore \dot{X}^\mu \dot{X}_\mu = \left(\frac{1}{l_s^2} p^2 + 2 \frac{1}{l_s} p^\mu \sum_n (\alpha_n^\mu e^{2in\sigma} + \tilde{\alpha}_n^\mu e^{-2in\sigma}) e^{-2in\tau} \right) + \dots$$

$$X'^\mu = -\frac{1}{l_s} \sum_n (\alpha_n^\mu e^{2in\sigma} - \tilde{\alpha}_n^\mu e^{-2in\sigma}) e^{-2in\tau}$$

$$X'^2 \cong \frac{1}{l_s^2} \sum_n \sum_m (\alpha_n^\mu \alpha_m^\mu e^{2i(n+m)\sigma} + \dots)$$

$$\therefore \text{we get terms such as } \int_0^l d\sigma \alpha_n^\mu (\alpha_m)_\mu e^{2i(n+m)\sigma} + \dots$$

$$\text{However we know that } \int d\sigma e^{2i(n+m)\sigma} = l \delta(n+m)$$

This is due to the fact that the length of the string

has compact support so we can extend the integral to $\int_{-\infty}^{\infty} d\sigma$.

$$* \boxed{H = \frac{1}{2} \sum_{-\infty}^{\infty} (\alpha_{-n} \cdot \alpha_n + \tilde{\alpha}_{-n} \cdot \tilde{\alpha}_n)} \quad , \quad \underline{\alpha_0^M := \tilde{\alpha}_0^M = \frac{1}{2} l_s p^M}$$

Open strings: $\boxed{H = \frac{1}{2} \sum_{-\infty}^{\infty} (\alpha_{-n} \cdot \alpha_n)}$, $\alpha_0^M = \tilde{\alpha}_0^M = l_s p^M$

Trivially H is conserved $\frac{d}{d\tau}(H) = 0$

* Stress-Energy tensor (Closed String):

In light-cone coordinates: $T_{--} = (\partial_- X_\mu)^2$, $T_{++} = (\partial_+ X_\mu)^2$

$$T_{+-} = T_{-+} = 0$$

$$\tau = (\sigma^+ + \sigma^-) \frac{1}{2}, \quad \sigma = (\sigma^+ - \sigma^-) \frac{1}{2}$$

Proof: $T_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} \eta_{\alpha\beta} \eta^{\mu\nu} \partial_\mu X^\nu \partial_\nu X_\mu = 0$

$$T_{++} = \partial_+ X^\mu \partial_+ X_\mu - \frac{1}{2} \eta_{++}^{\mu\nu} \partial_\mu X^\nu \partial_\nu X_\mu = 0$$

$$T_{+-} = \partial_+ X^\mu \partial_- X_\mu - \partial_- X^\mu \partial_+ X_\mu = 0 \quad = 0$$

$$X^\mu(\sigma, \tau) = X_R^\mu(\sigma^-) + X_L^\mu(\sigma^+)$$

$$\Rightarrow T_{++} = (\partial_+ X_L^\mu \partial_+ X_{L\mu}) = 0$$

$$T_{++} = (\partial_+ X_L^\mu \partial_+ X_{L\mu}) = 0$$

$$\begin{aligned} \nabla T_{++} &= \left(\frac{1}{2} l_s^2 p^M + l_s \sum_{n \neq 0} \tilde{\alpha}_n^M e^{-2in(\tau-\sigma)} \right) \\ &= l_s \alpha_0^M + l_s \sum_{n \neq 0} \tilde{\alpha}_n^M e^{-2in(\tau-\sigma)} \\ &= l_s^2 \left(\sum_{-\infty}^{\infty} \sum_{-\infty}^{\infty} \tilde{\alpha}_m^M \tilde{\alpha}_{m-n}^M e^{-2i(m+n)(\tau-\sigma)} \right) \end{aligned}$$

Let $m+n = p$

$$T_{++} = l_s^2 \sum_{-\infty}^{\infty} \sum_{-\infty}^{\infty} \tilde{\alpha}_{p-n}^M (\tilde{\alpha}_n)_M e^{-2ip(\tau-\sigma)}$$

we define

$$= 2 l_s^2 \sum_{-\infty}^{\infty} \tilde{L}_m e^{-2im(\tau-\sigma)}$$

$$\boxed{L_m := (2/2) \sum_{-\infty}^{\infty} \tilde{\alpha}_{m-n} \cdot \tilde{\alpha}_n} \quad (\text{Classical})$$

$$T_{--} = (\partial_- X_\mu)^2 \quad -2im(\tau-\sigma)$$

$$= 2\ell_s^2 \sum_{-\infty}^{\infty} L_m e$$

$$L_m = \frac{1}{2} \sum_{-\infty}^{\infty} \alpha_{m-n} \cdot \alpha_n \quad (\text{Classical})$$

• we will show that L_n are the conserved charges associated to conformal Killing fields.

